SAMPLE — SYNTHETIC FIELD DATA FOR DEMONSTRATION · NOT AN AQUATIC RESOURCES DELINEATION · NOT FOR REGULATORY SUBMITTAL

AQUATIC RESOURCES DELINEATION MAP SET

Sample Site — Fort Bend County, Texas · USACE Galveston District (SWG)

A white-label figure set and GIS package laid out against the Galveston District's GPS/GIS SOP (2016), its JD request checklist (July 2024) and the HQ Recommended Minimum Standards for Aquatic Resource Delineation Reports (July 2025), which SWG adopted and the Fort Worth District (SWF) hosts; the tables list where each item appears and what the client supplies. Real public data for a real private parcel (land-status checked); all field data are SYNTHETIC. Non-tidal site.

SHEET INDEX

Sheet	Figure	Title	Scale	Base
1	—	Cover and index	1:4,800 key map	NAIP 2022
2	1	Vicinity map, USGS topo, survey abstracts, Texas inset	1:24,000	USGS Topo (TNM)
3	2	Project area and property lines with measurements	1:2,400	NAIP 2022
4	3	Aerial imagery (2022, 2018, 1981-86), 1971 USGS map, 3DEP lidar	1:4,800	six panels
5	4	NRCS soils and hydric rating	1:4,800	NAIP 2022
6	5	National Wetlands Inventory and NHD	1:4,800	NAIP 2022
7	6	FEMA flood hazard zones (source disclosed)	1:12,000	USGS Topo (TNM)
8	7	Aquatic resources delineation (flags, data points, photos)	1:1,800	NAIP 2022

SUMMARY OF AQUATIC RESOURCES (SAMPLE — SYNTHETIC)

ID	Label	Cowardin	Acres	LF
W1	Wetland 1 (non-tidal)	PEM1A	0.6478	—
W2	Wetland 2 (non-tidal)	PEM1C	0.3249	—
W3	Wetland 3 (non-tidal)	PEM1A	0.1724	—
NWW1	Non-wetlands waters 1 (non-tidal)	R4SBCx	0.0715	621
	Upland (examined)		**25.0562**	
	Area not examined		**1.5807**	
	Project area		**27.8535**	

SWG / HQ ITEMS: WHERE THEY APPEAR IN THIS SET

Item (short)	Where
Map conventions, vicinity map (SOP d.i-ii)	every sheet; Sheet 2
Base layers: roads, hydro, county, NWI, imagery, DRG, lidar, FEMA, soils (d.iii)	Sheets 2-8
Reference block: project, surveyors, date (d.iv)	every sheet
Resource sizes; acreage not examined (d.v)	Sheet 8, Sheet 1
Numbered boundary points (d.vii); ESRI geodatabase in NAD83(2011)	Sheet 8; gis/GIS
Symbols told apart in colour and black and white (d.viii-ix)	all sheets (greyscale tests)
Point attributes: SOP item b, HQ GNSS guide fields (blank: synthetic)	gis/ (every point layer)
Property lines with measurements; parcel source and print date	Sheet 3
Solid bold project boundary with label; labels incl. "Upland"	Sheet 8
Texas state plane; NAD83 decimal degrees (6 dp); WGS 84 GeoPackage	all sheets; gis/
Survey abstract instead of S/T/R; flood source disclosed	Sheets 2, 7

FILES DELIVERED WITH THE SAMPLE

File	Content
SAMPLE_TX_Delineation_MapSet.pdf	8 sheets, 11 x 17 in, vector PDF (text searchable)
gis/GIS/*.shp (15 layers)	EPSG:6588 NAD83(2011) Texas South Central, US ft
gis/GIS/SWGSAMPLE_FortBend_NAD83_2011_20261010.gdb	ESRI file geodatabase, NAD83(2011) decimal degrees
gis/SWGSAMPLE_FortBend_WGS84_20261010.gpkg	GeoPackage, WGS 84 (HQ GNSS guide)
gis/GPS/*.csv	point exports, SOP item b and HQ GNSS fields (synthetic)
gis/SWGSAMPLE_FortBend_ORM_AquaticResources_NAD83_20261010.csv, .xlsx	ORM upload columns, NAD83(2011) (SOP g)
gis/*metadata*.txt	SOP metadata with a signature block (SOP e)
gis/SWGSAMPLE_FortBend_Delineation_EPSG6588_20261010.dxf	AutoCAD 2013 DXF; read back without errors in the ODA File Converter

KEY MAP — NAIP 2022-06-12, 1:4,800



GIS PACKAGE (gis/): GIS/ shapefiles in EPSG:6588 and an ESRI file geodatabase in NAD83(2011) decimal degrees (SOP); a WGS 84 GeoPackage (HQ GNSS guide); GPS/ point exports with the SOP item b and HQ GNSS fields; ORM aquatic resources table in NAD83(2011) (CSV and XLSX; ask the District for the current upload sheet); SOP metadata text with a signature block; AutoCAD 2013 DXF; read back without errors in the ODA File Converter. No Autodesk or Esri software was used.

CLIENT SUPPLIES: GNSS export (receiver, satellites, PDOP, dates, collector), wetland determination data forms (Atlantic and Gulf Coastal Plain Regional Supplement v2.0), OHWM data sheet, photographs, applicant name, SWG project number and the CAD property ID.
NOT IN THIS SAMPLE: jurisdictional labels (SWG checklist Table 2, only if the project manager asks: the HQ standard advises against them); MHW/HTL lines (non-tidal site); SWF-format permit drawings; a letter-size figure series; a land survey (RPLS).

COORDINATE SYSTEMS AND DATUMS

Data	System
Sheets, shapefiles, DXF	NAD83(2011) / Texas South Central (4204), US ft, EPSG:6588
Lat/long: sheets, geodatabase, ORM, GPS files	NAD83(2011) decimal degrees, 6 decimals (SOP)
GeoPackage (WGS 84)	null operation NAD83(2011) to WGS 84 (1); rigorous G2139 would move the site 1.2 m
Survey and pipeline layers (RRC)	NAD 27 sources, NADCON grids (0.35 m)
Vertical (contours, lidar)	NAVD88, US survey feet (3DEP)

AQUATIC RESOURCES DELINEATION MAP SET — TEXAS

Sample site · Fort Bend County, Texas · A-12, G. Borden Jr. Survey

Sample site — demonstration only (SYNTHETIC field data)

USACE Galveston District (SWG) GPS/GIS SOP (2016) · HQ ARDR standard (July 2025)

COVER AND INDEX

LEGEND

- Project area (27.85 ac)
- Wetland (synthetic)
- Area not examined

DATA SOURCES (queried by script; no live services in the package)

Layer	Source	Tier	Date
Parcels	Fort Bend CAD public parcel layer	T1	2026-10-09
Imagery	USDA NAIP (Planetary Computer STAC)	T1	2022-06-12
Wetlands	USFWS NWI (project Orchard, 1992 CIR)	T1	2026-10-10
Soils	NRCS SSURGO TX157 v.23 (SDA)	T1	2026-09-03
Lidar	USGS 3DEP, project TX_Houston_B24	T1	2024-03-29
Flood	FEMA DFIRM via Fort Bend County GIS; Esri NFHL copy	T2	2024-09-25
Surveys	Railroad Commission of Texas; FBCAD abstracts	T1	2026-10-10
Topo	USGS The National Map; HTMC 1971 scan	T1	2026-10-10
Hydrography	USGS NHD (large scale)	T1	2026-10-10
Land status	PAD-US 4.1, NCED, NRCS, TPWD, RIBITS	T1/T2	2026-10-10

REVISIONS

Rev.	Date	Change
0	2026-10-10	First issue of the Texas sample set
1	2026-10-10	Drafting refinements

TRUE NORTH

0 200 400 600 800 ft

Key map: 1 inch = 400 feet (1:4,800)

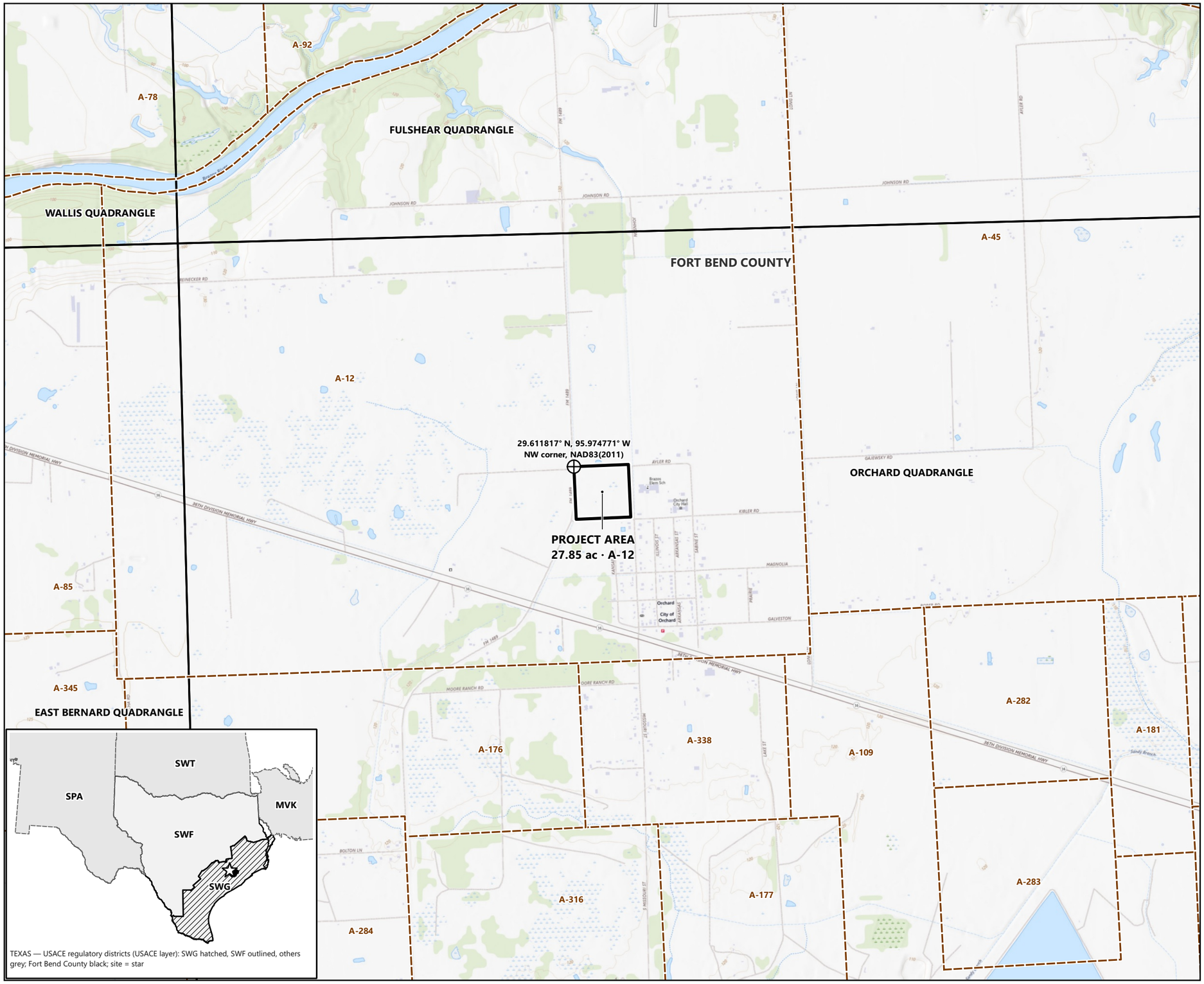
when printed at 11 × 17 in (full size)

Aquatic resource boundaries, flags, data points and photo points are SYNTHETIC (placed on lidar depressions, the NWI and the roadside ditch for demonstration). No field delineation was performed. No jurisdictional status is shown (HQ standard, July 2025).
Land status: no record found on or within 0.5 mi of the project area in PAD-US 4.1, NCED, NRCS easements (NEST), TPWD WMA / state park / Farm and Ranch Lands layers or RIBITS (EPA copy, 2021) (queried 2026-10-10; positive control run for every layer).
Property lines: Fort Bend Central Appraisal District public parcel layer (FBCAD Public Data, ArcGIS Online; data edited 2026-10-09; geometry only, queried 2026-10-10; printed 2026-10-10). Lengths measured on that geometry. Approximate relative location of property boundaries, not an on-the-ground survey. CAD property ID withheld in this sample: [client inserts CAD ID].
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

Project: Sample Site, Fort Bend County, Texas · SWG project no.: [client]
Applicant: [client name] (sample) · Surveyors: none (field data SYNTHETIC)
Map creator: Danir Kantserov — GIS/CAD mapping (independent contractor, white-label sample)
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TEXAS — USACE regulatory districts (USACE layer): SWG hatched, SWF outlined, others grey; Fort Bend County black; site = star

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USACE Galveston District (SWG) GPS/GIS SOP (2016) · HQ ARDR standard (July 2025)

VICINITY MAP — USGS TOPO 1:24,000 (FIGURE 1)

LEGEND

- Project area (27.85 ac)
- Original Texas land survey line (RRC survey layer; abstract no.)
- USGS 7.5-minute quadrangle boundary and name

SITE LOCATION

Item	Value
County, state	Fort Bend County, Texas
USACE district	Galveston (SWG); USACE Regulatory Boundary layer
Original Texas Land Survey	Abstract 12, Gail Borden Jr. Survey (RRC; FBCAD A-12)
USGS 7.5′ quadrangle	Orchard, Tex. (cell 29095-E8)
Centre (NAD83(2011))	29.610352° N, 95.972978° W
State plane zone	Texas South Central (4204), EPSG:6588, US ft
Directions	North-east corner of FM 1489 and Kibler Road, 2,495 ft (0.5 mi) NNW of the Orchard place point (OpenStreetMap)

CORNER COORDINATES (NAD83(2011); EPSG:6588)

Corner	Lat. (N)	Long. (W)	X (ft)	Y (ft)
NW	29.611817	95.974771	2,929,630.1	13,782,429.1
NE	29.611861	95.971244	2,930,750.0	13,782,474.3
SE	29.608889	95.971182	2,930,797.7	13,781,394.2
SW	29.608839	95.974709	2,929,677.8	13,781,347.1

TRUE NORTH

0 1,000 2,000 3,000 4,000 ft

Scale: 1 inch = 2,000 feet (1:24,000)

when printed at 11 × 17 in (full size)

Base: USGS Topo, The National Map (basemap.nationalmap.gov USGSTopo service, as served 2026-10-10), exported at 1:24,000.
Survey lines and abstract numbers: Railroad Commission of Texas public viewer, Surveys layer (source NAD 27), requested in NAD 27 geographic and transformed to NAD83(2011) with the NADCON grids (PROJ "NAD27 to NAD83 (7)", 0.35 m); RRC survey data are an alternative to the GLO's OTLS grid and may differ from it. Second source at the site: FBCAD abstract layer (A-12, Gail Borden).
Texas has no PLSS: the abstract and survey name replace section, township and range (HQ standard: S/T/R "recommended if available").
The whole frame lies in Fort Bend County (RRC Counties layer); see the inset. Quadrangle grid: USGS map indices (carto.nationalmap.gov).
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

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Sample site · Fort Bend County, Texas · A-12, G. Borden Jr. Survey

Sample site — demonstration only (SYNTHETIC field data)

USACE Galveston District (SWG) GPS/GIS SOP (2016) · HQ ARDR standard (July 2025)

PROJECT AREA AND PROPERTY LINES ON NAIP 2022 (FIGURE 2)

LEGEND

- Project area (27.85 ac, entire FBCAD parcel)
- Adjoining property lines (FBCAD, approximate; IDs withheld)
- Gas gathering pipeline, abandoned (RRC, approximate)
- Gas gathering pipeline, in service (RRC, approximate)
- Ditch / stream line (USGS NHD, FCode 33600 = canal/ditch)
- Existing fenced gravel pad, oil and gas equipment (NAIP 2022)

PROPERTY CORNERS (NAD83(2011); EPSG:6588)

Vertex	X (ft)	Y (ft)	Lat. (N)	Long. (W)
V1	2,929,677.8	13,781,347.1	29.608839	95.974709
V2	2,929,632.7	13,782,349.7	29.611598	95.974770
V3	2,929,630.1	13,782,429.1	29.611817	95.974771
V4	2,930,750.0	13,782,474.3	29.611861	95.971244
V5	2,930,752.5	13,782,418.2	29.611707	95.971241
V6	2,930,797.7	13,781,394.2	29.608889	95.971182

PROPERTY LINES WITH MEASUREMENTS

Line	Grid bearing	Length (ft)
V1–V2	N 02°34′ W	1,003.6
V2–V3	N 01°52′ W	79.5
V3–V4	N 87°42′ E	1,120.8
V4–V5	S 02°32′ E	56.1
V5–V6	S 02°32′ E	1,025.0
V6–V1	S 87°36′ W	1,120.8

TRUE NORTH

0 100 200 300 400 ft

Scale: 1 inch = 200 feet (1:2,400)

when printed at 11 × 17 in (full size)

Property lines: Fort Bend Central Appraisal District public parcel layer (FBCAD Public Data, ArcGIS Online; data edited 2026-10-09; geometry only, queried 2026-10-10; printed 2026-10-10). Lengths measured on that geometry. Approximate relative location of property boundaries, not an on-the-ground survey. CAD property ID withheld in this sample: [client inserts CAD ID].
Parcel size: 27.8535 ac from the FBCAD polygon (planar, EPSG:6588); CAD record 30 ac. FBCAD x,y (EPSG:2278, NAD83) used as served (shift to NAD83(2011) about 0.6 m).
Pipelines: RRC Pipelines layer (T-4 data; NAD 27 by NADCON), approximate. Ditches: USGS NHD. Aerial: NAIP 2022-06-12 (0.6 m; Planetary Computer STAC).
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

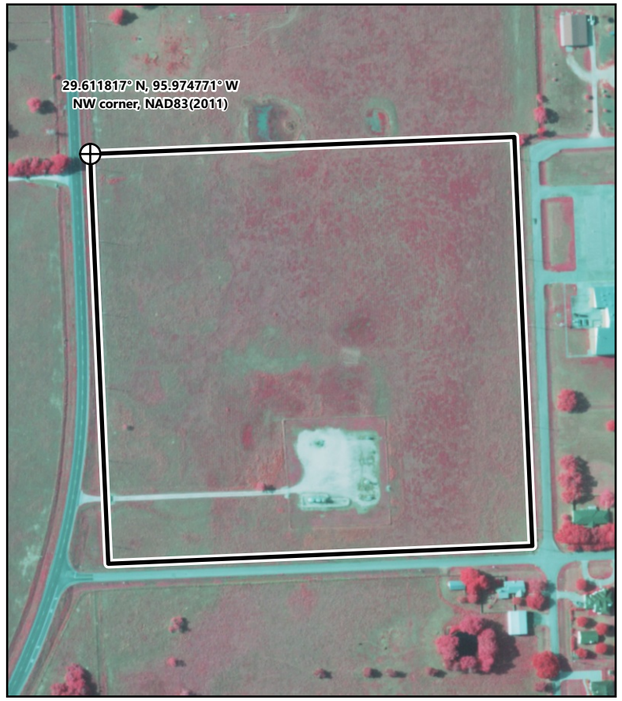
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A — NAIP 2022-06-12, natural colour, 0.6 m (USDA FSA; Planetary Computer STAC).



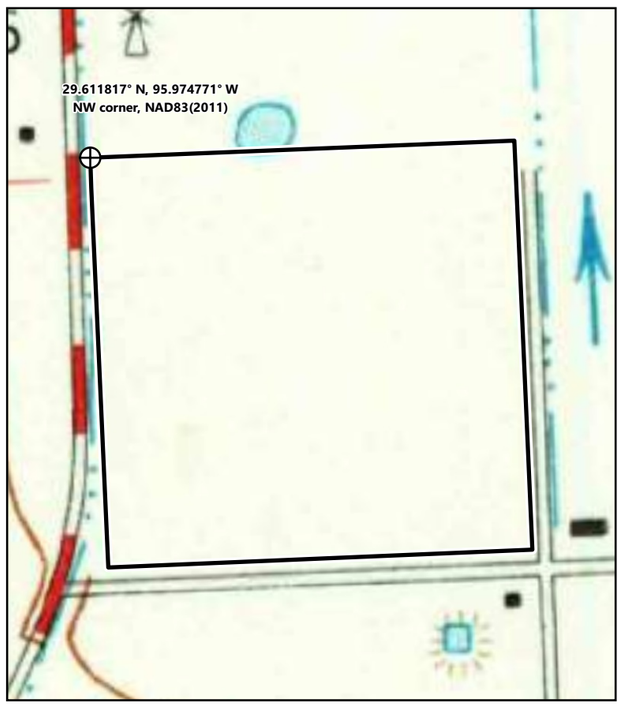
B — NAIP 2022-06-12, colour infrared (bands NIR, red, green). Same flight as A.



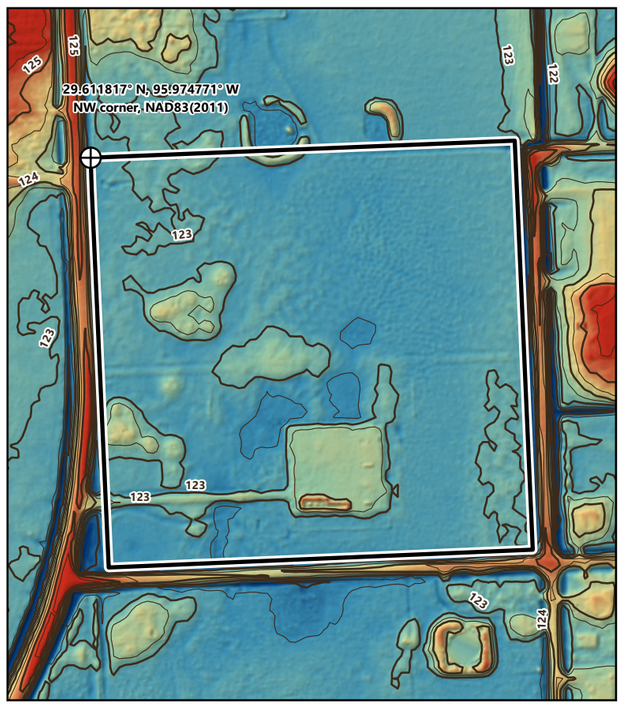
C — NAIP 2018-12-05 (December), natural colour, 0.6 m.



D — NHAP colour infrared, flown January 1981 to May 1986 (TxGIO record; frame date not given).



E — USGS 1:24,000 Orchard, Tex. quadrangle, 1971 (HTMC scan 116077; NAD 27, reprojected).



F — 3DEP lidar TX_Houston_B24 (QL1, flown 2024-02-18 to 2024-03-29), 0.5-ft contours, NAVD88 ft.

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USACE Galveston District (SWG) GPS/GIS SOP (2016) · HQ ARDR standard (July 2025)

AERIAL IMAGERY, HISTORICAL TOPO AND LIDAR (FIGURE 3)

LEGEND

Project area (27.85 ac)

Contours, NAVD88 ft (3DEP 1-m DEM averaged to 15 ft)

1-ft contour (contours under 250 ft long omitted)

0.5-ft contour

PANELS

Panel	Date	Source	Resolution
A	2022-06-12	NAIP natural colour	0.6 m
B	2022-06-12	NAIP colour infrared	0.6 m
C	2018-12-05	NAIP natural colour	0.6 m
D	1981-1986	NHAP colour infrared	0.75 m
E	1971	USGS 7.5' topographic map	1:24,000
F	2024-02/03	3DEP lidar (QL1)	1 m

SOURCE ITEMS (traceable to the cache)

Panel	Source item
A, B	NAIP STAC item tx_m_2909525_nw_15_060_20220612
C	NAIP STAC item tx_m_2909525_nw_15_060_20181205_20190428
D	TxGIO image service NHAP/NHAP81_CIR_75cm; collection e732df59
E	HTMC GeoTIFF TX_Orchard_116077_1971_24000_geo.tif
F	3DEP dynamic service, raster IDs 17637,63379 (TX_Houston_B24)

Panel F elevation tint, feet NAVD88 (shaded by the hillshade):

121.5 · 122.5 · 123.5 · 124.5 · 125.5

TRUE NORTH

0 200 400 600 800 ft

Scale (all panels): 1 inch = 400 feet (1:4,800)

when printed at 11 × 17 in (full size)

Scale: panels A-F and Figures 4 and 5 are at 1:4,800, the scale of the cover key map (HQ standard: environmental-data maps at the same scale, to the extent practicable). Figure 6 (flood) is at 1:12,000 to show the nearest special flood hazard area.
Range of conditions: June 2022 (A, B), December 2018 (C) and an earlier NHAP image (D, January 1981 to May 1986; frame date not given). Antecedent precipitation (APT) not evaluated: no image was relied on to delineate (synthetic data).
Historical topography: USGS Historical Topographic Map Collection (TNM API), the SOP's "DRG" layer.
Lidar: USGS 3DEP, raster locked to TX_Houston_B24 (QL1, USGS Lidar Base Specification 2024 Revision A); tint 121.5 to 125.5 ft.
NAIP producer: USDA Farm Service Agency (Planetary Computer STAC record). NHAP: TxGIO collection "Texas NHAP Imagery"; record: acquired between January 1981 and May 1986.
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

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NRCS SOILS AND HYDRIC RATING (FIGURE 4)

LEGEND

- Project area (27.85 ac, entire FBCAD parcel)
- Map unit, hydric rating: predominantly hydric (66-99 percent)
- Map unit, hydric rating: predominantly nonhydric (1-32 percent)
- Map unit, hydric rating: nonhydric (0 percent)

SOIL MAP UNITS IN THE PROJECT AREA

Unit	Map unit name	Acres	%	Hydric %	Components (percent)
Ea	Edna loam, 0 to 1 percent slopes	18.6573	67.0	5 (predominantly nonhydric)	Edna 87; Cieno 5 (hydric); Bernard 5; Telferner 3
Bb	Bernard clay loam, 0 to 1 percent slopes	9.1962	33.0	1 (predominantly nonhydric)	Bernard 90; Lake Charles 5; Edna 2; Verland 2; Leton 1 (hydric)
	Total = project area	**27.8535**	**100.0**		

MAP UNIT PROPERTIES (SDA muaggatt, component)

Unit	Drainage class (dominant)	Hydrologic group	Hydric components
Ea	Somewhat poorly drained	D	Cieno 5%
Bb	Somewhat poorly drained	D	Leton 1%

MAP UNITS IN THE FRAME: HYDRIC RATING (5 CLASSES)

Unit	Map unit name (SSURGO)	Hydric % (class)
Bb	Bernard clay loam, 0 to 1 percent slopes	1 (predominantly nonhydric)
Be	Bernard-Edna complex, 0 to 1 percent slopes	2 (predominantly nonhydric)
Ea	Edna loam, 0 to 1 percent slopes	5 (predominantly nonhydric)
KraA	Krasna fine sandy loam, 0 to 2 percent slopes	0 (nonhydric)
La	Lake Charles clay, 0 to 1 percent slopes	1 (predominantly nonhydric)
Wa	Gessner loam, 0 to 1 percent slopes, frequently ponded	98 (predominantly hydric)

TRUE NORTH

0 200 400 600 800 ft

Scale: 1 inch = 400 feet (1:4,800)

when printed at 11 × 17 in (full size)

Soils: NRCS SSURGO, Fort Bend County, Texas (TX157), survey version 23, established 2026-09-03; polygons from the Soil Data Access WFS, components and ratings from SDA tables (queried 2026-10-10). Hydric rating by map unit (5 classes) = percent of the map unit in components rated hydric; computed from the SDA component table and equal to SDA muaggatt.hydclprs for every unit.
Acres: planar (EPSG:6588), rounded by largest remainder so that the rows add up to the project area.
Imagery: USDA NAIP, natural colour, 0.6 m, acquired 2022-06-12 (quarter-quad tx_m_2909525_nw_15_060_20220612), from the Microsoft Planetary Computer STAC (public). Most recent NAIP served as a public image service; NAIP 2024 exists for Texas but no service was found.
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

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NATIONAL WETLANDS INVENTORY AND NHD (FIGURE 5)

LEGEND

- Project area (27.85 ac, entire FBCAD parcel)
- NWI freshwater emergent wetland
- NWI freshwater pond
- NWI riverine (incl. ditches)
- USGS NHD flowline (canal/ditch, stream)
- USGS NHD waterbody

NWI POLYGONS IN THE PROJECT AREA

NWI code	Meaning (FWS code diagram)	Acres in project	Polygon acres
PEM1A	Palustrine, Emergent, Persistent, Temporarily Flooded	0.6408	0.6408
R5UBFx	Riverine, subsystem 5 (not defined in the FGDC 2013 standard or the 2019 FWS code diagram), Unconsolidated Bottom, Semipermanently Flooded, Excavated	0.1755	3.0339
PEM1C	Palustrine, Emergent, Persistent, Seasonally Flooded	0.0981	0.0981
PUSCx	Palustrine, Unconsolidated Shore, Seasonally Flooded, Excavated	0.0269	0.3137

OTHER NWI CODES IN THE FRAME

NWI code	Meaning
PEM1Cx	Palustrine, Emergent, Persistent, Seasonally Flooded, Excavated
PUBHx	Palustrine, Unconsolidated Bottom, Permanently Flooded, Excavated
R4SBC	Riverine, Intermittent, Streambed, Seasonally Flooded

TRUE NORTH

0 200 400 600 800 ft

Scale: 1 inch = 400 feet (1:4,800)

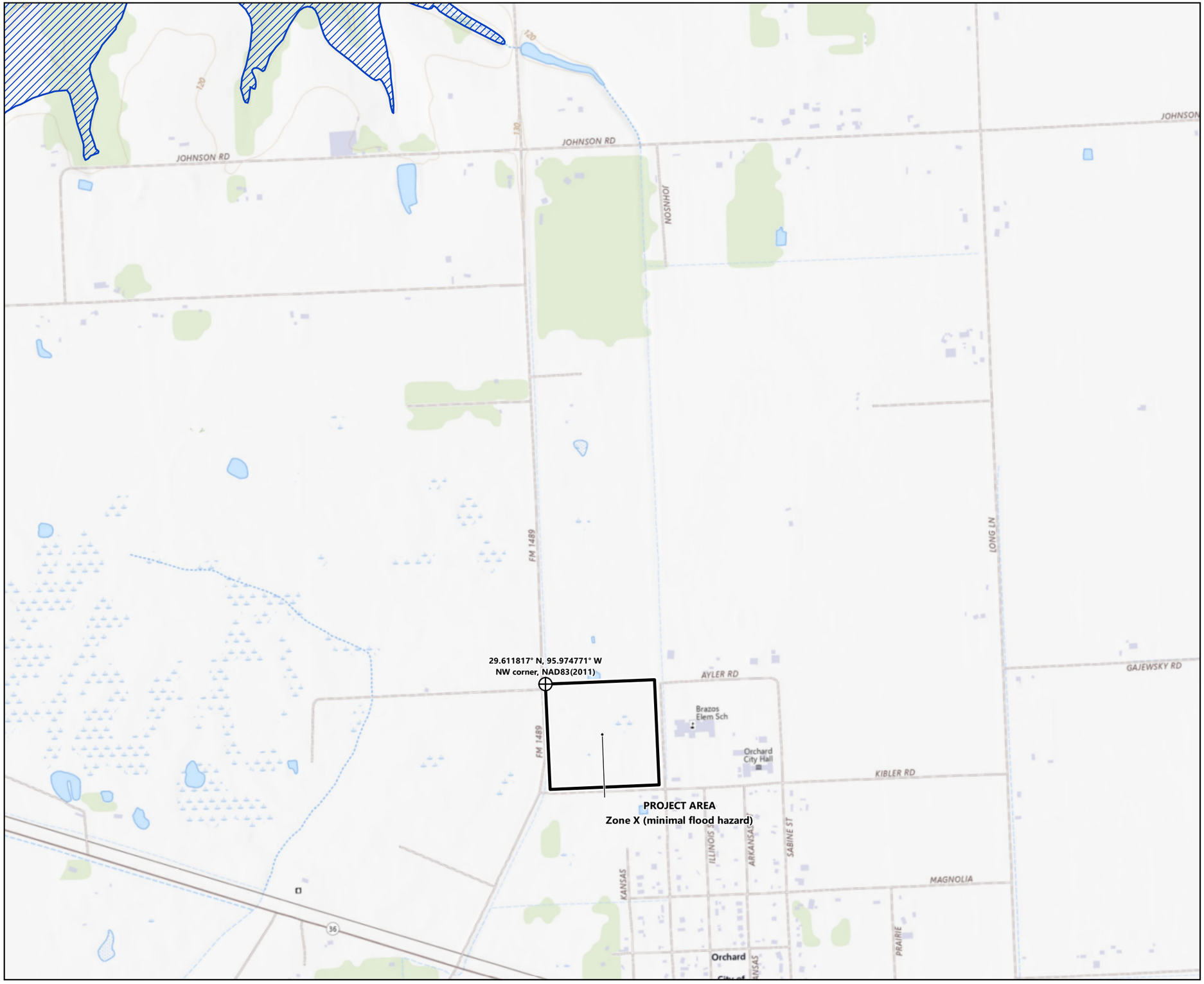
when printed at 11 × 17 in (full size)

NWI: USFWS National Wetlands Inventory Wetlands layer (fwspublicservices.wim.usgs.gov, queried 2026-10-10). NWI project at the site: "Orchard", CIR imagery 11/92 (1992), 1:65,000 (USFWS Data_Source layer).
Codes decoded with the FWS code diagram (Feb 2019) and the FGDC classification standard (2013); subsystem 5 (R5) is defined in neither. NWI is a desktop inventory: the delineation (Figure 7) differs from it, as field delineations do.
NHD: USGS National Hydrography Dataset, large-scale flowlines and waterbodies (hydro.nationalmap.gov). Imagery: USDA NAIP, natural colour, 0.6 m, acquired 2022-06-12 (quarter-quad tx_m_2909525_nw_15_060_20220612), from the Microsoft Planetary Computer STAC (public). Most recent NAIP served as a public image service; NAIP 2024 exists for Texas but no service was found.
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

Project: Sample Site, Fort Bend County, Texas · SWG project no.: [client]
Applicant: [client name] (sample) · Surveyors: none (field data SYNTHETIC)
Map creator: Danir Kantserov — GIS/CAD mapping (independent contractor, white-label sample)
Created 2026-10-10 · Rev. 1 (2026-10-10) · revised by: Danir Kantserov
Map file: SAMPLE_TX_Delineation_MapSet.pdf

SAMPLE — not for regulatory submittal

SAMPLE — SYNTHETIC FIELD DATA FOR DEMONSTRATION · NOT AN AQUATIC RESOURCES DELINEATION · NOT FOR REGULATORY SUBMITTAL



AQUATIC RESOURCES DELINEATION MAP SET — TEXAS

Sample site · Fort Bend County, Texas · A-12, G. Borden Jr. Survey

Sample site — demonstration only (SYNTHETIC field data)

USACE Galveston District (SWG) GPS/GIS SOP (2016) · HQ ARDR standard (July 2025)

FEMA FLOOD HAZARD ZONES 1:12,000 (FIGURE 6)

LEGEND

- Project area (27.85 ac)
- Zone AE (1-percent annual chance, base flood elevations)
- Zone X, area of minimal flood hazard (outside the zones above)

FLOOD HAZARD SUMMARY

Item	Value
Zone at the site	X, area of minimal flood hazard (27.8535 ac, 100 percent)
Special flood hazard area on site	none (0 ac)
Nearest SFHA	Zone AE, 6,070 ft NNW (edge to edge; bearing 341° from the parcel centre)
DFIRM ID	48157C (DFIRM_ID field of both copies)
FIRM panel / effective date	not confirmed (see below); confirm at msc.fema.gov

FLOOD SOURCES COMPARED

Source	Data date	Zone at site	Nearest SFHA
Fort Bend County GIS copy (drawn)	2024-09-25	X (minimal)	6,070 ft
Esri copy of the NFHL (cross-check)	NFHL of 2025-05-07	none mapped	6,070 ft

FIRM PANEL

FIRM panel number and effective date not confirmed (FEMA services not reached from our network on 2026-10-10; no other panel index found); confirm at msc.fema.gov.

TRUE NORTH

0 500 1,000 1,500 2,000 ft

Scale: 1 inch = 1,000 feet (1:12,000)

when printed at 11 × 17 in (full size)

Flood zones: FEMA DFIRM 48157C flood hazard areas as published by Fort Bend County GIS ("FBC Flood Plain Zone 2021 (view)", ArcGIS Online; data edited 2024-09-25). Site 100 percent in Zone X, area of minimal flood hazard. Zone X polygons of different FEMA studies merged for display (kept separate in the package). No regulatory floodway lies in the frame (nearest 8,675 ft NNW).
Cross-check: Esri copy of FEMA's NFHL (May 7, 2025; omits Zone X minimal, D, NP and areas not included): no flood hazard area on the site; nearest special flood hazard area 6,070 ft (Zone AE, NNW) in both copies.
Base: USGS Topo (The National Map). Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).

Project: Sample Site, Fort Bend County, Texas · SWG project no.: [client]
Applicant: [client name] (sample) · Surveyors: none (field data SYNTHETIC)
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AQUATIC RESOURCES (SAMPLE — SYNTHETIC)

ID	Label	Cowardin	Acres	Sq ft	LF
W1	Wetland 1 (non-tidal)	PEM1A	0.6478	28,217	—
W2	Wetland 2 (non-tidal)	PEM1C	0.3249	14,155	—
W3	Wetland 3 (non-tidal)	PEM1A	0.1724	7,511	—
NWW1	Non-wetlands waters 1 (non-tidal)	R4SBCx	0.0715	3,113	621
	Wetlands, total		**1.1451**		
	Non-wetlands waters, total		**0.0715**		**621**
	Upland (examined)		**25.0562**		
	Area not examined		**1.5807**		
	Project area		**27.8535**		

DATA POINTS — FORM NO. = DP NO. (SYNTHETIC)

Point	Form	Plot	Resource	Lat. (N)	Long. (W)
DP-1	1	wetland	Wetland 1	29.610515	95.972483
DP-2	2	upland	Wetland 1	29.610385	95.972892
DP-3	3	wetland	Wetland 2	29.609851	95.973389
DP-4	4	upland	Wetland 2	29.610181	95.973248
DP-5	5	wetland	Wetland 3	29.610414	95.974035
DP-6	6	upland	Wetland 3	29.610562	95.973984
DP-7	7	upland	check: swale	29.608950	95.973811
DP-8	8	upland	check: NWI PUSCx	29.611768	95.973232
DP-9	9	upland	check: Bb unit	29.610339	95.971830
OH-1	OHWM 1	OHWM	Non-wetl. waters 1	29.610987	95.974738

PHOTO POINTS (photographs not included)

Photo	View (true)	Subject
P1	61°	Wetland 1 (non-tidal)
P2	196°	Wetland 2 (non-tidal)
P3	181°	Wetland 3 (non-tidal)
P4	355°	NWW1, looking north along the ditch
P5	184°	NWW1, looking south along the ditch
P6	92°	Area not examined (fenced pad)
P7	91°	Upland swale along the south boundary
P8	1°	Upland pasture (Bernard clay loam)

FLAGS: 50 boundary flags with IDs printed (W1-1 … NWW1-7), numbered clockwise from the northernmost flag of each wetland and south to north along the ditch (east OHWM). Spacing between consecutive wetland flags 28.7–45.0 ft (median 41.7 ft). NWW1 is mapped inside the project area only: south of NWW1-1 (the southern 419 ft of the west line) the lidar ditch lies wholly west of the parcel line, in the FM 1489 right-of-way. SOP item b and HQ GNSS fields on every point layer of the GIS package. Nearest NGS mark: AW4704 "A 804 RESET" (bench mark disk; horizontal position scaled, no published height), about 1,200 ft from the centre. Nearest mark with an adjusted NAD 83(2011) position and a published height: DR8909 "PAM 62 ARP", 1.2 mi.

AERIAL: USDA NAIP, acquired 2022-06-12 (0.6 m, natural colour)
Microsoft Planetary Computer STAC (public)

AQUATIC RESOURCES DELINEATION MAP SET — TEXAS

Sample site · Fort Bend County, Texas · A-12, G. Borden Jr. Survey

Sample site — demonstration only (SYNTHETIC field data)

USACE Galveston District (SWG) GPS/GIS SOP (2016) · HQ ARDR standard (July 2025)

AQUATIC RESOURCES DELINEATION ON NAIP 2022 (FIGURE 7)

LEGEND

- Project area (27.85 ac, entire FBCAD parcel)
- Wetland, PEM (boundary flagged, synthetic)
- Non-wetlands waters (roadside ditch; east OHWM flagged, west edge = project line)
- Area not examined for aquatic resources (1.5807 ac)
- Boundary flag with printed ID (synthetic; 50 flags)
- Data point, wetland plot (label = data form no.)
- Data point, upland plot (label = data form no.)
- OHWM data sheet point (OH-1)
- Photo point P# and view direction
- Geographic reference point (NAD83(2011) lat/long)
- Adjoining property lines (FBCAD parcel layer, approximate)

DELINEATION SUMMARY (SYNTHETIC)

Project area: 27.8535 ac (entire FBCAD parcel; examined 26.2728 ac)
Wetlands: 3, 1.1451 ac · Non-wetlands waters: 1, 0.0715 ac, 621 LF
Upland: 25.0562 ac · Area not examined: 1.5807 ac
Data points: 9 (forms 1–9) + OH-1 · Photo points: 8
Boundary flags: 50, IDs printed on the map

TRUE NORTH

0 50 100 150 200 ft

Scale: 1 inch = 150 feet (1:1,800)
when printed at 11 × 17 in (full size)

Aquatic resource boundaries, flags, data points and photo points are SYNTHETIC (placed on lidar depressions, the NWI and the roadside ditch for demonstration). No field delineation was performed. No jurisdictional status is shown (HQ standard, July 2025).
Property lines: Fort Bend Central Appraisal District public parcel layer (FBCAD Public Data, ArcGIS Online; data edited 2026-10-09; geometry only, queried 2026-10-10; printed 2026-10-10). Lengths measured on that geometry. Approximate relative location of property boundaries, not an on-the-ground survey. CAD property ID withheld in this sample: [client inserts CAD ID].
Land status: no record found on or within 0.5 mi of the project area in PAD-US 4.1, NCED, NRCS easements (NEST), TPWD WMA / state park / Farm and Ranch Lands layers or RIBITS (EPA copy, 2021) (queried 2026-10-10; positive control run for every layer).
Coordinates: NAD83(2011) / Texas South Central zone, US survey feet (EPSG:6588, verified in the QGIS/PROJ database). Latitude/longitude in decimal degrees, NAD83(2011) (6 decimals), as in the geodatabase, the ORM table and the GPS files. The GeoPackage is WGS 84 by the null operation NAD83(2011) to WGS 84 (1).
Areas: planar, from the GIS geometry (EPSG:6588); acres rounded once to 4 decimals; totals are sums of the rows shown.
Original Texas Land Survey: Abstract 12, Borden, G Jr. (RRC survey layer, NAD 27 transformed by NADCON; FBCAD abstract layer: A-12, Gail Borden).
Labels: SWG JD checklist Table 1 ("Wetland X (non-tidal)", "Non-wetlands waters X (non-tidal)", "Upland"); resource type + number per HQ (July 2025). Photo views: true azimuths (the arrows are drawn on the grid; convergence 1.48° here). Non-tidal site: no NWI polygon of the estuarine or marine system or the riverine tidal subsystem within about 25 km (25 tiles of 0.1 degree, counted 2026-10-10).

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